

TrES-5b

R-0882

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250828 · created 2026-07-29T00:48:53+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460915.7142 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.052
Transit depth (Rp/Rs)^2	3.08 +/- 1.82 [%]
Orbital Inclination (inc)	83.32 +/- 2.92
Airmass coefficient 1 (a1)	69.81 +/- 33.12
Airmass coefficient 2 (a2)	-3.0 +/- 1.67
Scatter in the residuals of the lightcurve fit is	40.88 %
Best Comparison Star	#3 - [590, 82]
Optimal Aperture	2.42
Optimal Annulus	12.36
Transit Duration (day)	0.044 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.052 vs 0.142 (deviation 0.63σ)
Duration fit vs published	0.044 d vs 0.0758 d (deviation 1.99σ)
Mid-time O–C	9.4 min
Depth SNR	3.4
Residual scatter	40.88 %

Light curve

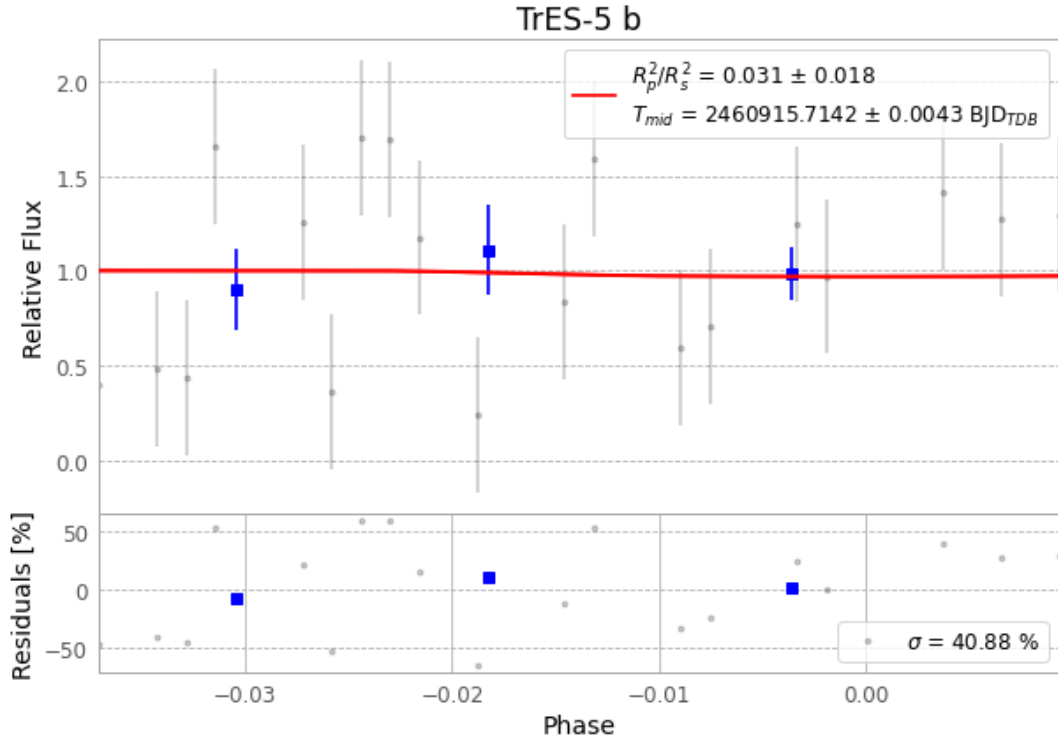


Figure 1 — FinalLightCurve_TrES-5 b_2025-08-27.png (SHA-256 d9d83e84d5c58c63...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [613, 456], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e1b173581c69c5ae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

36 FITS frames (24 MB), TRES-5250828034215.FITS → TRES-5250828052715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250828.log (ee56228dbd72...), FinalLightCurve_TrES-5 b_2025-08-27.png (d9d83e84d5c5...), FinalLightCurve_TrES-5 b_2025-08-27.csv (d08e71b9d05e...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 00:48Z.